

Fake News Detection Capstone Report

1. Introduction

Fake news and online misinformation have become major issues in modern society. In this project, machine learning and deep learning techniques were used to classify news articles as real or fake based on their textual content. The goal was to compare different models and evaluate their effectiveness for fake news detection.

2. Dataset Description

The dataset consisted of two CSV files containing fake and real news articles. After combining and shuffling the datasets, the final dataset contained 44,898 articles with relatively balanced classes. The dataset included article titles, article text, subject, date, and labels indicating whether the article was fake or real.

3. Models Used

Three machine learning models were trained and compared: Logistic Regression, Random Forest, and Linear SVM. The text data was cleaned and converted into numerical features using TF-IDF vectorization before training. Random Forest achieved the highest performance with an F1-score of approximately 0.998, while all models achieved very high accuracy overall.

4. Neural Network

A neural network was built using Keras Dense layers with ReLU activation and Dropout layers to reduce overfitting. The network was trained for 10 epochs using the same TF-IDF features from the machine learning models. The neural network achieved validation accuracy above 99% with only minimal overfitting.

5. Results & Comparison

All models performed extremely well on the fake news classification task. Random Forest slightly outperformed the other models and achieved the best overall accuracy and F1-score. Although the neural network also achieved very strong performance, the classical machine learning models performed slightly better on the TF-IDF text data.

6. What I Learned

This project demonstrated the importance of text preprocessing and feature extraction in NLP tasks. Cleaning the text and using TF-IDF significantly improved model performance. I also learned that more complex models, such as neural networks, are not always better than traditional machine learning models for structured text classification problems.

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